Welcome to CSC331 Fall 2025 Data Structures

Professor Kevin Byron (kbyron@bmcc.cuny.edu)
Department of CIS, Room F-1030-I
Office hours: Mon 2pm-5pm (in-person and Zoom)
Office phone: 212-220-1490
Borough of Manhattan Community College/CUNY

CSC331 Spring 2025
Data Structures

Week 7:

Graph definitions
Prim's MST algorithm

Professor Kevin Byron
Department of CIS
Borough of Manhattan Community College/CUNY

Reminders

- No class Mon 10-20-2025
- Follow Monday schedule on: Fri 10-24-2025
- Program 3 due date
  - 1159pm Fri 10-31-2025
  - late submission receives no credit
- Exam 2
  - Section 501 Tue 11-11-2025 – Zoom 630pm-730pm
  - Section 172 Wed 11-12-2025 – In person room M-1001 630pm-730pm
    - Closed book, closed notes, no devices
  - Section 500 Wed 11-12-2025 – In person room N-453 1010am-1110am
    - Closed book, closed notes, no devices
- Program 4 due date
  - 1159pm Fri 11-21-2025
  - late submission receives no credit
- Posted on Brightspace
  - Week 7 lecture slides

Graphs

- Sample graphs
  - A=(Va,Ea):
    - Va={1,2,3,4,5}
    - Ea={(1,2),(1,3),(1,4),(2,5),(3,4),(3,5)}
  - B=(Vb,Eb):
    - Vb={1,2,3,4,5,6}
    - Eb={(1,2),(1,3),(2,4),(2,5),(3,6)}
  - C=(Vc,Ec):
    - Vc={Chicago,LA,Miami,NY,Omaha}
    - Ec={(Chicago,LA),(Chicago,Miami),(Chicago,NY),(Chicago,Omaha),(LA,NY),(LA,Omaha),(Miami,NY)}

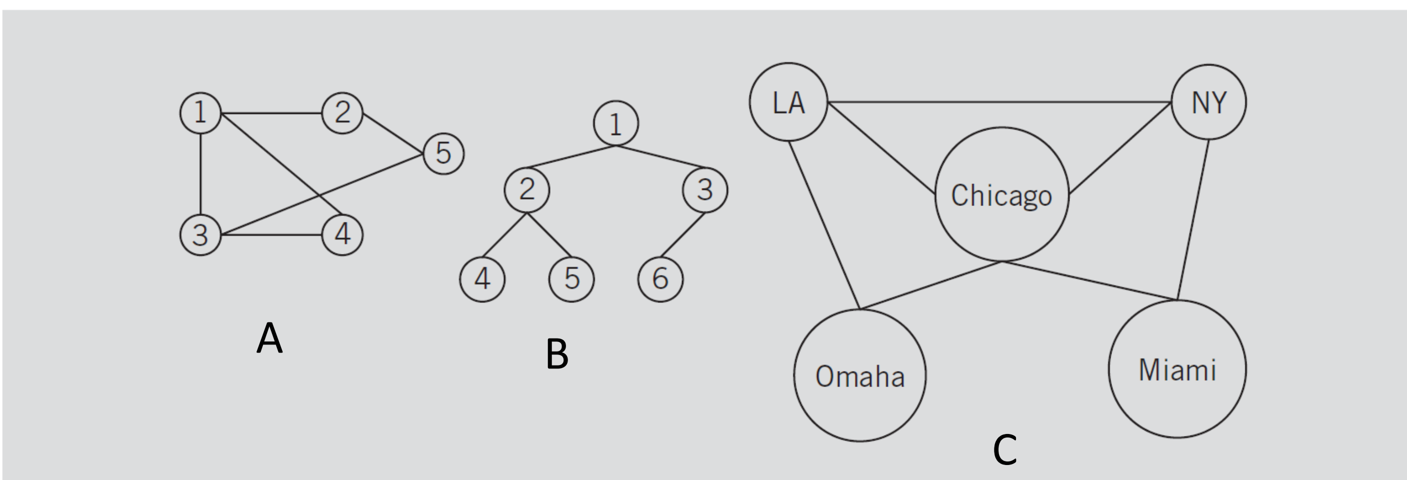


FIGURE 12-3 Various undirected graphs

Graphs

- Sample graphs

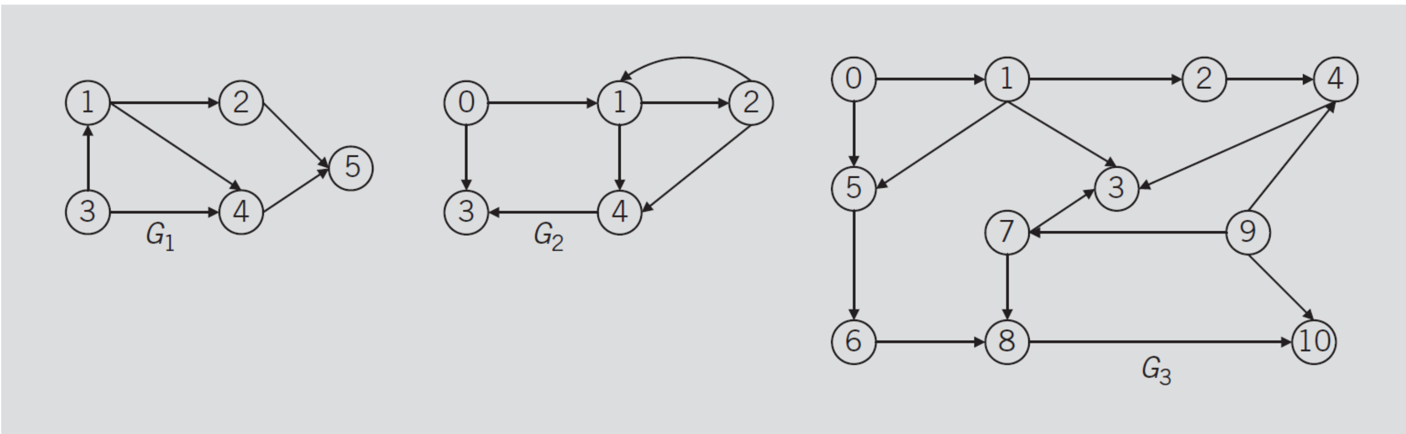


FIGURE 12-4 Various directed graphs

For the graphs of Figure 12-4, we have

$V(G_1) = \{1, 2, 3, 4, 5\}$ $E(G_1) = \{(1, 2), (1, 4), (2, 5), (3, 1), (3, 4), (4, 5)\}$
$V(G_2) = \{0, 1, 2, 3, 4\}$ $E(G_2) = \{(0, 1), (0, 3), (1, 2), (1, 4), (2, 1), (2, 4), (4, 3)\}$
$V(G_3) = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ $E(G_3) = \{(0, 1), (0, 5), (1, 2), (1, 3), (1, 5), (2, 4), (4, 3), (5, 6), (6, 8), (7, 3), (7, 8), (8, 10), (9, 4), (9, 7), (9, 10)\}$

Graphs

- Graph representation: adjacency matrix

Consider the directed graphs of Figure 12-4. The adjacency matrices of the directed graphs G_1 and G_2 are as follows:

$$A_{G_1} = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},\ A_{G_2} = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Graphs

- Graph representation: adjacency list

Consider the directed graphs of Figure 12-4. Figure 12-5 shows the adjacency list of the directed graph G_2.

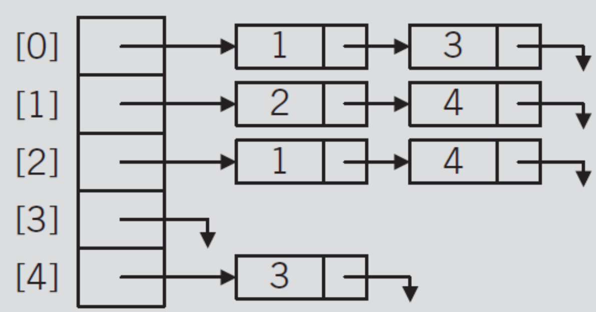
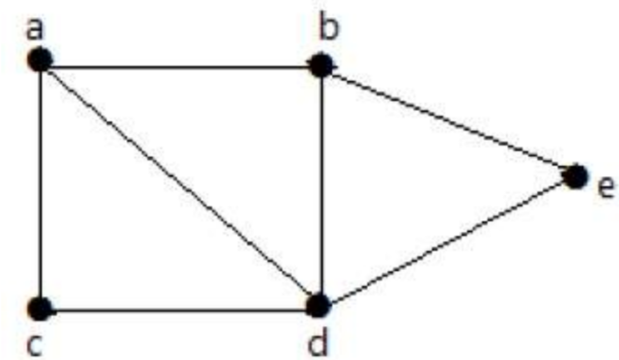


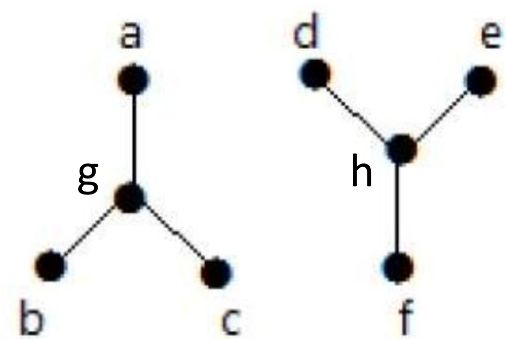
FIGURE 12-5 Adjacency list of graph G_2 of Figure 12-4

Graphs

- A graph is said to be connected if there is a path (i.e., set of edges) between every pair of vertices
- In a connected graph, from every vertex to any other vertex, there should be some path to traverse
- Sample connected graph, J=(Vj,Ej):
  - Vj={a,b,c,d,e}
  - Ej={(a,b),(a,c),(a,d),(b,d),(b,e),(c,d),(d,e)}

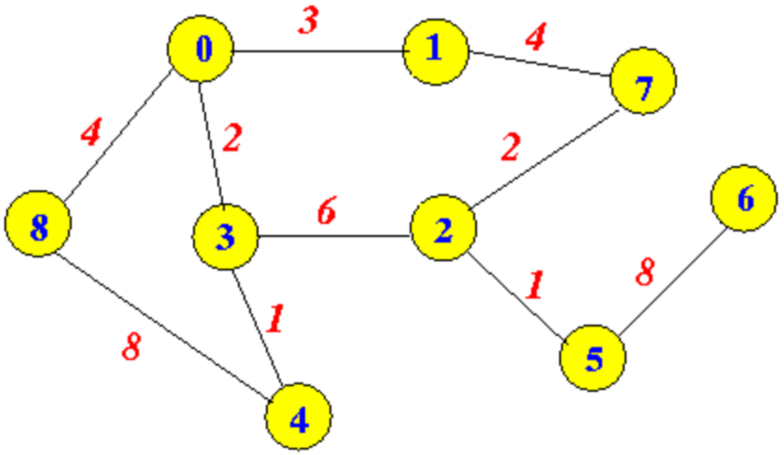


- Sample unconnected graph, K=(Vk,Ek):
  - Vk={a,b,c,d,e,f,g,h}
  - Ek={(a,g),(b,g),(c,g),(d,h),(e,h),(f,h)}



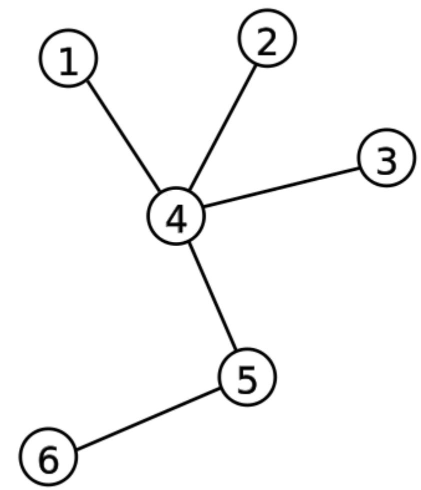
Graphs

- A graph is said to be weighted if each edge of has an associated numerical value, called a weight
- A weight may represent a value such as cost, distance, etc.
- A weighted graph may be directed or undirected
- Sample weighted graph, G=(Vg,Eg):
  - Vg={0,1,2,3,4,5,6,7,8}
  - Eg={(0,1,3),(0,3,2),(0,8,4),(1,7,4),(2,3,6),(2,5,1),(3,4,1),(4,8,8),(5,6,8)}



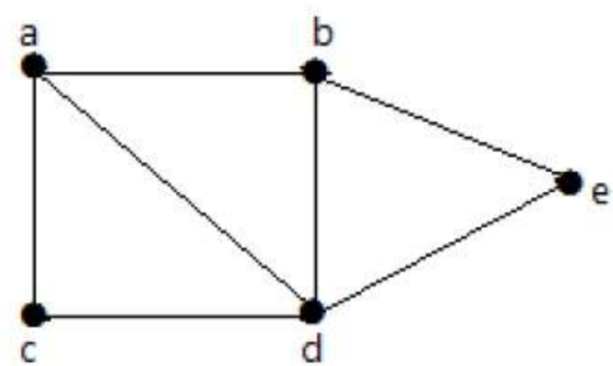
Graphs

- A graph is said to be a tree if any two vertices are connected by exactly one path
- Every acyclic connected graph is a tree, and vice versa.
- A tree with n vertices has (n-1) edges
- Sample tree, T=(Vt,Et):
  - Vt={1,2,3,4,5,6}
  - Et={(1,4),(2,4),(3,4),(4,5),(5,6)}
- In a tree, one vertex may be named the root



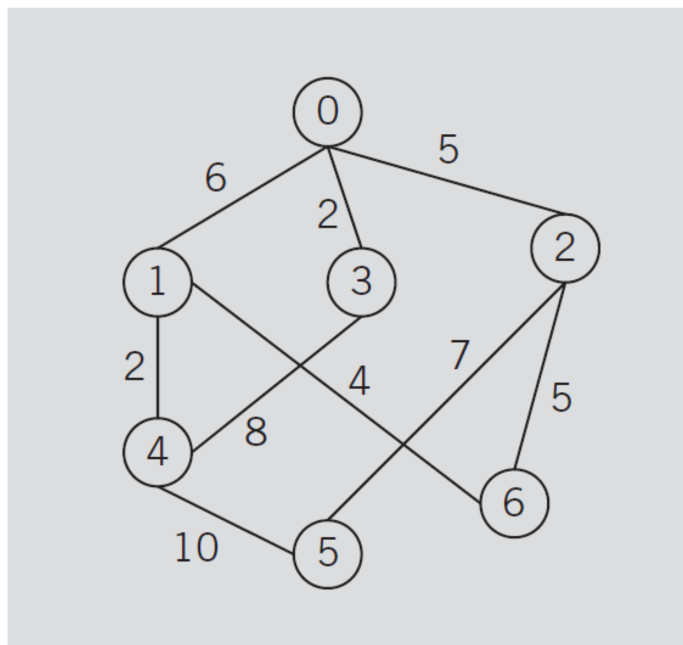
Graphs

- A subgraph of a graph G is another graph, H, formed from a subset of the vertices and edges of G
- The vertex subset must include all endpoints of the edge subset
- A spanning subgraph is a subgraph that contains all the vertices of the original graph
- A spanning tree is a spanning subgraph that is also a tree
- Find a spanning subgraph, Qs={Vs,Es), and a spanning tree, Qt=(Vt,Et), of this graph:, Q={Vq,Eq)



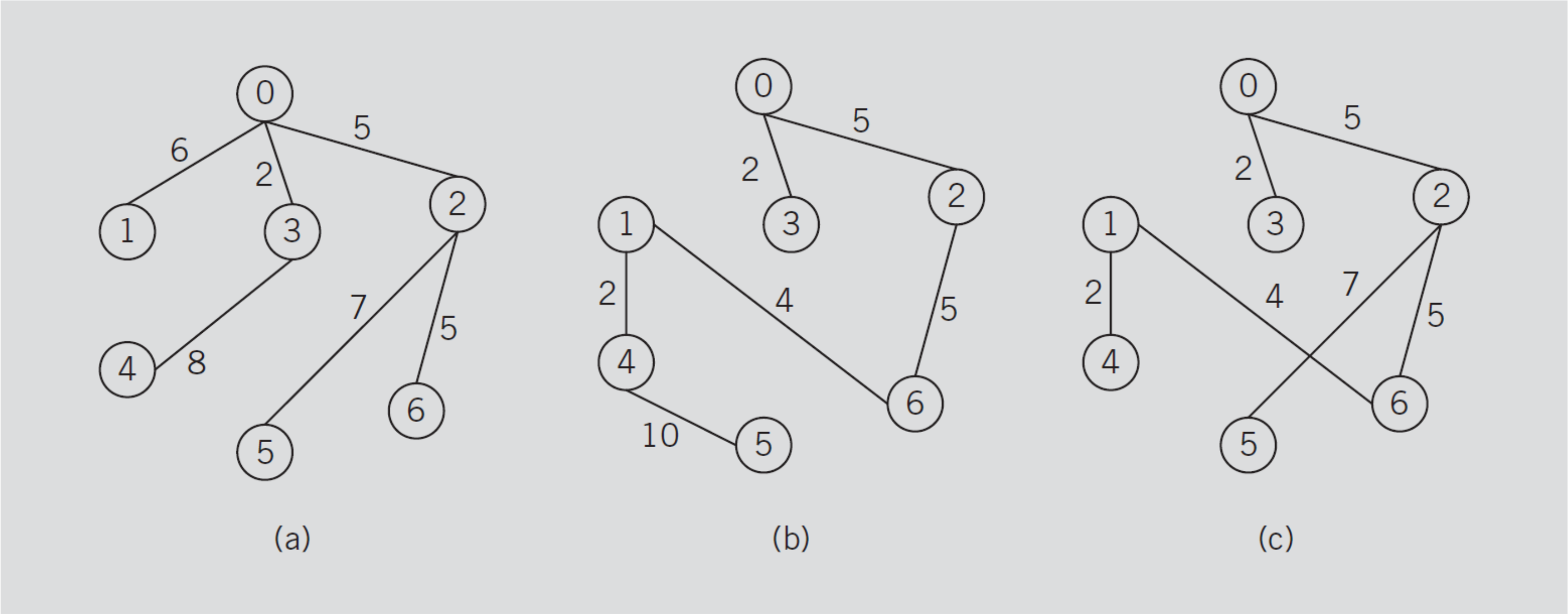
Graphs

- A minimum spanning tree (MST) is a spanning tree of a connected, weighted graph and which has the minimum total edge weight among all spanning trees of the original graph
- Find the minimum spanning tree of this graph:



Graphs

- Spanning trees from the graph on the previous slide:



Graphs

- Prim's MST algorithm:

The general form of Prim's algorithm is as follows. (Let n be the number of vertices in G.)

1. Set V(T) = {source}
2. Set E(T) = empty
3. for i = 1 to n

    3.1. minWeight = infinity;

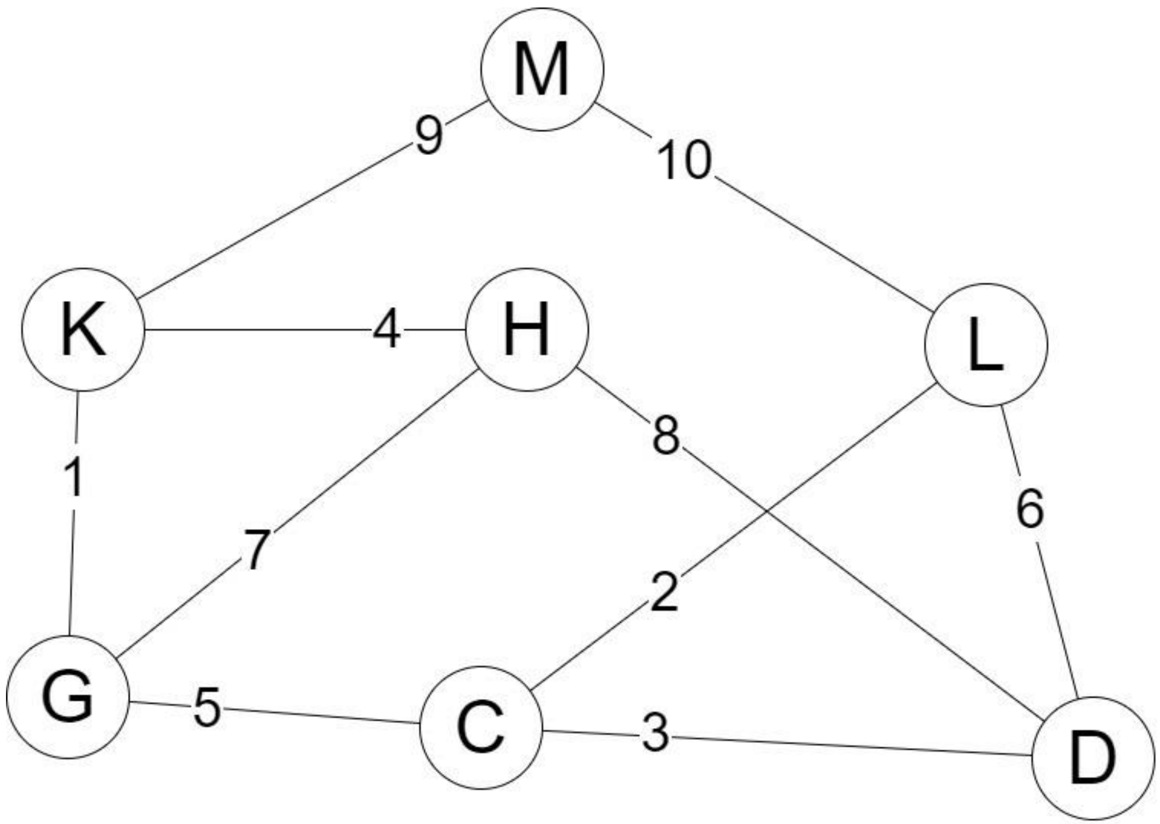
    3.2.
```
for j = 1 to n
    if vj is in V(T)
      for k = 1 to n
        if vk is not in T and weight[vj, vk] < minWeight
        {
            endVertex = vk;
            edge = (vj, vk);
            minWeight = weight[vj, vk];
        }
```

    3.3. V(T) = V(T) ∪ {endVertex};

    3.4. E(T) = E(T) ∪ {edge};

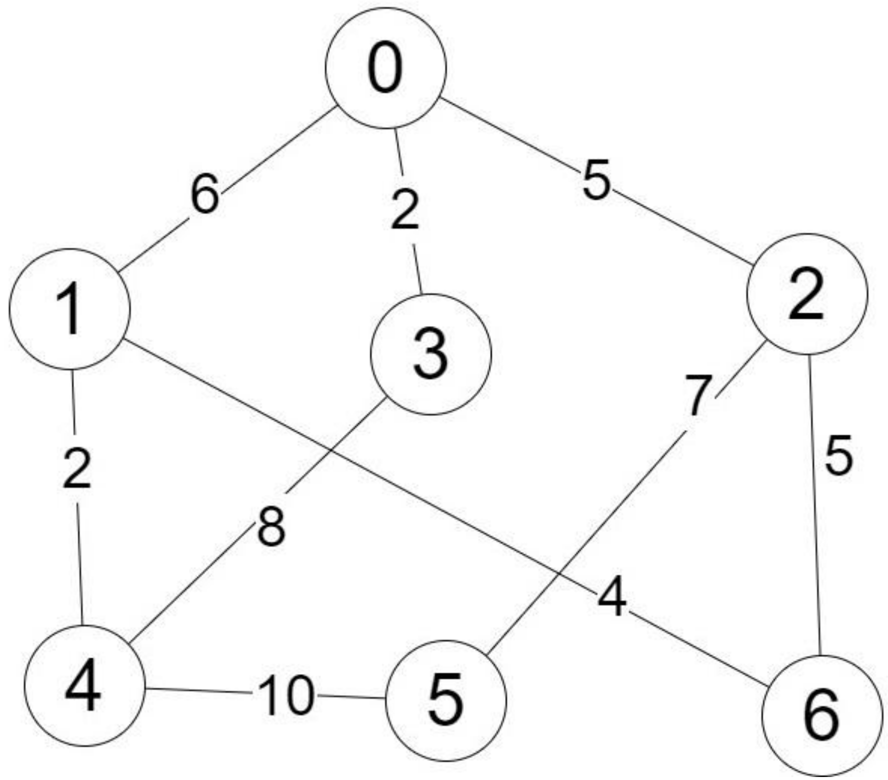
Graphs

- Prim’s MST drill D1:
  - Start with node C … identify the first 4 edges collected …



Graphs

- Prim's MST drill D2:
  - Start with node 6 … identify the first 4 edges collected …



Assignment

- Read Malik chapter 11 – binary trees
- Read Malik chapter 12 - graphs